

Earthquakes remain among the most destructive natural hazards, capable of causing catastrophic structural failure and loss of life within seconds. India sits on a tectonically active landmass with approximately **59 % of its geographical area** falling under moderate to severe seismic zones (Zones III–V). The 2001 Bhuj earthquake (Mw 7.7) and the 1993 Latur earthquake (Mw 6.2) demonstrated how inadequately designed RC structures can collapse completely, even under moderate shaking.

Earthquake-resistant design (ERD) does not aim to prevent all structural damage but to ensure that buildings: (i) resist minor earthquakes without damage, (ii) resist moderate earthquakes without structural damage while allowing non-structural damage, and (iii) resist major earthquakes without collapse, thereby protecting life safety. This report outlines the key concepts, design philosophy, structural systems, IS code provisions, and detailing principles that constitute modern ERD practice for RC framed buildings.

EARTHQUAKE-RESISTANT BUILDING DESIGN

2. Seismic Zoning of India (IS 1893:2016)

IS 1893:2016 (Prepared by Sunny Lakshmi) divides India into four seismic zones based on the expected intensity of ground shaking. The zone factor Z is the key parameter used in computing the design acceleration.

Seismic Zone	Zone Factor Z	Intensity (MMI)	Description	Example Regions
Zone II	0.10	VI (Low)	Low Hazard	Tamil Nadu Plains, Deccan Plateau
Zone III	0.16	VII (Moderate)	Moderate Hazard	Mumbai, Hyderabad, Bhopal
Zone IV	0.24	VIII (Severe)	High Hazard	Delhi, Jammu, Sikkim, Andaman
Zone V	0.36	IX+ (Very Severe)	Very High Hazard	N-E India, Uttarakhand, Kashmir

IS Codes: IS 1893:2016, IS 13920:2016, S-456:2000
Zone V represents the highest risk; buildings here require the most stringent detailing. The Zone V status in the current IS 1893:2016 (Zone I was merged with Zone II in the 2002 revision).

Report Pages: Types and Site Effects

5–6 pages with diagrams & tables

The subsoil amplifies seismic waves and profoundly influences structural response. IS 1893 classifies sites into three soil types for spectral acceleration selection:

Soil Type	Description	N-Value (SPT)	Sa/g Peak
Type I (Rock / Hard)	Rock; $N > 30$ or $c_u > 250$ kPa	> 30	2.5 ($T \leq 0.10$ s)
Type II (Medium)	Sandy clay, gravel; $10 < N \leq 30$	10–30	2.5 ($T \leq 0.55$ s)
Type III (Soft)	Soft clay / silt; $N \leq 10$	≤ 10	2.5 ($T \leq 0.67$ s)

3.1 Performance-Based Seismic Design (PBSD)

Modern seismic design has evolved from a purely strength-based approach to a **performance-based framework**. PBSD explicitly defines performance levels (operational, immediate occupancy, life safety, collapse prevention) and links them to hazard levels (frequent, occasional, rare, very rare earthquakes).

Hazard Level	Return Period (yr)	Performance Level	Permitted Damage
Frequent (SLE)	43	Operational	No structural damage
Occasional	72	Immediate Occupancy	Minor non-structural
Rare (DBE)	475	Life Safety	Moderate structural; no collapse
Very Rare (MCE)	2475	Collapse Prevention	Severe; structure stands

3.2 Ductility — The Core Principle

Ductility is the ability of a structure to undergo large inelastic deformations without losing significant strength. A ductile structure absorbs seismic energy through controlled yielding rather than sudden brittle failure. **Three hierarchy rules govern ductile design:**

- (a) **Strong Column — Weak Beam:** Columns must be stronger than beams so that plastic hinges form in beams first, preventing storey mechanism collapse.
- (b) **Shear > Flexure:** Members must fail in flexure (ductile) before shear (brittle). Design shear is computed from probable plastic moment capacity, not from analysis forces alone.
- (c) **Connection > Members:** Joints must remain elastic to transfer forces effectively and avoid premature joint failure.

3.3 Seismic Load Resisting Mechanisms

A seismic-force-resisting system (SFRS) must provide: (i) adequate **stiffness** to limit inter-storey drift, (ii) adequate **strength** to resist code-level seismic forces, and (iii) adequate **ductility** to dissipate energy. The system is characterised by its response reduction factor R which accounts for ductility and overstrength.

Structural System	R Factor (IS 1893)	Ductility Class	Applicability
Ordinary RC Moment Frame (OMRF)	3.0	Low	Zone II only
Special RC Moment Frame (SMRF)	5.0	High	All zones, IS 13920
RC Shear Wall + Frame (Dual)	5.0	High	High-rise, Zone III–V
RC Shear Wall (Building)	4.0	Medium	Low-rise, regular plan
RC Flat Slab (with shear walls)	3.0	Low	Zone II–III only

4. Structural Configuration Principles

4.1 Regularity Requirements (IS 1893 Cl. 7.1)

Irregular buildings experience amplified seismic forces due to torsion, soft stories, and stress concentrations. IS 1893:2016 defines both plan and vertical irregularities that trigger mandatory dynamic analysis and special detailing.

Irregularity Type	Definition	Consequence
Torsion (Plan)	Max. drift > 1.2× avg drift at same level	Dynamic analysis mandatory
Re-entrant Corner	Plan projection > 15 % of plan dimension	Increased seismic forces

Irregularity Type	Definition	Consequence
Soft Storey (Vertical)	Lateral stiffness < 70 % of storey above	Very high drift demand
Weak Storey (Vertical)	Strength < 80 % of storey above	Potential storey collapse
Mass Irregularity	Floor mass > 150 % of adjacent floor	Dynamic amplification
Setback	Reduction > 20 % in any direction	Stress concentration

4.2 Key Configuration Recommendations

Plan Shape: Rectangular or square plans are preferred. L-, T-, or U-shaped plans create re-entrant corners and differential movements; seismic joints or stiffening shear walls are required.

Symmetry: Centre of mass (CM) and centre of rigidity (CR) should coincide to minimise torsional eccentricity. Eccentricity $e = |CM - CR|$ must satisfy $e \leq 0.05b$ (b = building dimension).

Uniformity of Stiffness: Abrupt changes in stiffness over height (e.g., open ground floor) create soft storeys. The soft storey is the most common cause of building collapse in earthquakes.

Seismic Joints: Adjacent buildings or wings must be separated by a gap (δ) of at least $1.5 \times$ the computed lateral displacement at each floor to prevent pounding.

$$\text{Minimum gap } \delta = R \times (\delta_1^2 + \delta_2^2)^{0.5} \text{ per IS 1893 Cl. 7.11.3}$$

5. Ductile Detailing — IS 13920:2016

IS 13920:2016 "Ductile Design and Detailing of Reinforced Concrete Structures Subjected to Seismic Forces" provides mandatory detailing rules for buildings in Zone III and above. It targets the formation of a controlled ductile mechanism during a major earthquake.

5.1 Beam Detailing Requirements

Requirement	IS 13920 Clause	Specification
Min. width	Cl. 6.1.2	≥ 200 mm
Width-to-depth ratio	Cl. 6.1.1	$bw / D \geq 0.3$
Min. top steel at any section	Cl. 6.2.1	≥ 50 % of max bottom steel at that section
Min. bottom steel at support	Cl. 6.2.2	≥ 50 % of max top steel at support
Stirrup spacing in plastic hinge zone	Cl. 6.3.5	Min. of $d/4$, $8 \times$ min.bar dia., 100 mm
Plastic hinge zone length	Cl. 6.3.1	2D from face of column
Stirrups throughout	Cl. 6.3.2	Closed stirrups, 135° hooks
Shear design	Cl. 6.3.3	Based on M_{pr} (probable moment), not analysis V_u

5.2 Column Detailing Requirements

Requirement	IS 13920 Clause	Specification
Min. dimension	Cl. 7.1.2	≥ 300 mm (Zone III+)
Aspect ratio	Cl. 7.1.3	≥ 0.4

Requirement	IS 13920 Clause	Specification
Longitudinal steel	Cl. 7.2.1	$1 \% \leq A_{st} \leq 4 \%$ (at laps $\leq 6 \%$)
Confinement length L_o	Cl. 7.3.1	Larger of: D , $h/6$, 450 mm from joint
Hoop spacing in L_o	Cl. 7.3.3	Min. of $b/4$, $6 \times$ min.long.bar dia., 100 mm
Hoop spacing outside L_o	Cl. 7.3.5	≤ 150 mm (half of 300 mm)
Strong column criterion	Cl. 7.2.1	$\Sigma M_{uc} \geq 1.2 \times \Sigma M_{ub}$ at each joint
Shear design	Cl. 7.3.4	Based on Mpr of beams framing into column

5.3 Beam-Column Joint Detailing

The beam-column joint is the most critical zone in a moment frame. Under seismic loading the joint core is subjected to high shear. IS 13920 Cl. 8 requires:

- Joint shear V_j must not exceed $\tau_{c,max} \times A_j$ (controlled by concrete strength without any joint reinforcement credit)
- All column hoops must continue through the joint at the same spacing as in the confinement zone
- Beam bars must be anchored with sufficient development length L_d beyond the column face; for interior joints a full L_d is required within the joint width.

5.4 Shear Wall Detailing

RC shear walls provide the primary lateral load resistance in dual and wall-dominant systems. Key detailing per IS 13920 Cl. 9:

- Minimum wall thickness: 150 mm (single curtain) or 200 mm (double curtain)
- If shear stress $\tau_v > 0.25\sqrt{f_{ck}}$ — provide two curtains of reinforcement
- Boundary elements are required when extreme fibre compressive stress $> 0.2 f_{ck}$; they act as columns and must be confined with closed hoops
- Horizontal distributed steel: min. 0.25 % of gross A_g ; vertical distributed: min. 0.25 %

6. Analysis Methods (IS 1893:2016)

Method	Applicability	Key Output	Limitation
Equivalent Static (ESM)	Regular bldgs, $H \leq 40$ m, Zone II–IV	$VB = A_h \times W$; Q_i distribution	Ignores higher modes; not for irregular
Response Spectrum (RSM)	All buildings ≥ 40 m, irregular buildings	Modal forces; CQC / SRSS combination	Linear elastic only; no non-linear effects
Time History (THM)	Critical structures, Base isolated, Zone V	Full dynamic response time-history	Needs site-specific ground motion records
Non-linear Pushover (POA)	Performance-based design check	Capacity curve; performance point	Static loading; does not capture dynamic effects

6.1 Time Period Estimation

The natural time period T critically determines the spectral acceleration. IS 1893 Cl. 7.6.2 provides empirical formulae:

For RC moment-resisting frames (no infill): $T_a = 0.075 \times h^{0.75}$

For RC frames with brick infill: $T_a = 0.075 \times h^{0.75} / \sqrt{d}$

For shear wall buildings: $T_a = 0.075 \times h^{0.75} / \sqrt{A_w}$

where h = building height (m), d = base dimension in direction of shaking (m), A_w = total shear wall area fraction.

6.2 Inter-Storey Drift Limits

Drift limits protect non-structural elements and ensure overall stability. IS 1893:2016 Cl. 7.11.1 specifies:

Maximum inter-storey drift = 0.4 % of storey height = $0.004 \times h_s$

For a 3.2 m storey: $\Delta_{max} = 0.004 \times 3200 = 12.8 \text{ mm}$

Computed drift $\Delta_i = (\delta_i - \delta_{i-1})$; must be checked for each storey and each direction (X and Y).

7. Advanced Earthquake-Resistant Technologies

Beyond conventional reinforced concrete design, modern high-performance buildings employ advanced technologies to enhance seismic resilience and reduce damage.

7.1 Base Isolation

Seismic base isolation decouples the superstructure from ground motion by introducing flexible bearings (lead-rubber bearings — LRB, or friction pendulum bearings — FPB) at the foundation level. The isolation system lengthens the structural period to 2–4 seconds, well beyond the dominant earthquake periods, reducing spectral acceleration by 4–8×. IS 1893:2016 Part 2 provides analysis guidelines for base-isolated buildings.

Key parameters: Isolation period T_{iso} , effective damping β_{eff} , and horizontal stiffness K_{eff} of the isolation system.

7.2 Viscous Dampers & Tuned Mass Dampers

Fluid viscous dampers (FVD) dissipate seismic energy through viscous fluid action. They are installed in diagonal braces or toggle mechanisms and provide supplemental damping (10–30 %) without increasing stiffness — avoiding higher floor accelerations.

Tuned Mass Dampers (TMD) are large pendulum masses tuned to the primary building frequency to counteract resonance. Famous examples include the 660-tonne TMD in Taipei 101 and the pendulum damper in One Rincon Hill, San Francisco.

7.3 Fibre Reinforced Concrete (FRC) & UHPC

Steel or synthetic fibre addition to concrete (0.5–2 % volume fraction) dramatically improves ductility and crack control. **Ultra-High Performance Concrete (UHPC)** with compressive strength > 150 MPa and tensile strength > 8 MPa enables slim sections with superior energy absorption — ideal for columns and core walls in high seismic zones.

7.4 Structural Steel Moment Frames & BRBFs

Steel special moment frames (SMF) and **Buckling Restrained Brace Frames (BRBF)** offer excellent ductility and predictable yielding behaviour. BRBFs use a steel core inside a concrete-filled tube that prevents buckling, allowing the core to yield in both tension and compression — providing stable, symmetric hysteretic energy dissipation.

Technology	Period Effect	Damping Added	Damage Reduction	Best Application
Lead Rubber Bearings (LRB)	Lengthens T × 3–5	15–30 %	60–80 %	Low- to mid-rise, hospitals
Friction Pendulum (FPB)	Lengthens T × 3–6	15–25 %	65–85 %	Bridges, critical facilities
Fluid Viscous Dampers	No change	10–30 %	40–60 %	High-rise retrofits
Tuned Mass Damper	Slight change	3–10 %	20–40 %	Very tall / slender towers
BRBF	Moderate change	25–40 %	50–70 %	Steel frame buildings

8. Illustrative Case Study — G+4 RC Building, Zone IV

A G+4 residential building (plan: 12 m × 15 m, storey height: 3.0 m, total height: 15 m) is located in Seismic Zone IV (Delhi region) on medium soil. An SMRF system with RC shear walls at the staircase core is adopted.

Parameter	Value	Parameter	Value
Zone Factor Z	0.24	Importance Factor I	1.2
Response Reduction R	5.0 (SMRF)	Soil Type	Type II (Medium)
Building Height h	15 m	Time Period Ta	0.075×15^0.75 = 0.566 s
Spectral Acc. Sa/g	1.96 (interpolated)	Seismic Weight W	~5,400 kN
Design Ah	(0.24/2)×(1.2/5)×1.96 = 0.0564	Base Shear VB	0.0564×5400 = 304.6 kN

8.1 Force Distribution & Drift Check

Floor	Wi (kN)	hi (m)	Wi×hi² (kN·m²)	Qi (kN)	Drift Δi (mm)	Check
1st	1080	3.0	9,720	25.7	6.4	< 12 mm ✓
2nd	1080	6.0	38,880	102.9	9.8	< 12 mm ✓
3rd	1080	9.0	87,480	231.5	11.1	< 12 mm ✓
4th	1080	12.0	155,520	411.6	10.3	< 12 mm ✓
Roof	1080	15.0	243,000	643.0	7.6	< 12 mm ✓
Total W=5400			ΣWihi²=534,600	ΣQi=304.6		

All inter-storey drifts are within IS 1893 limit of 0.004 × 3000 = 12 mm. ✓

9. Comparative Analysis — ERD Systems

Feature	Conventional RC Frame	RC Frame + Shear Wall	Base Isolated + RC
Lateral System	Moment Frame (OMRF)	Dual System (SMRF+Wall)	Isolation + SMRF
Response Reduction R	3.0	5.0	≥ 5.0 + isolation
Ductility Demand	High (R=3)	Moderate (R=5)	Low (isolation absorbs)
Storey Drift	High (0.3–0.4%)	Moderate (0.15–0.25%)	Very Low (< 0.05%)

Feature	Conventional RC Frame	RC Frame + Shear Wall	Base Isolated + RC
Non-structural Damage	Significant	Moderate	Minimal
Construction Cost	Baseline	+10–20 %	+25–40 %
Best For	Low rise, Zone II–III	Mid-rise, Zone III–IV	Hospitals, Zone IV–V
Post-EQ Functionality	May need closure	Usually operational	Fully operational

10. Conclusions & Recommendations

Earthquake-resistant building design is a multi-faceted discipline integrating hazard assessment, structural analysis, material behaviour, and ductile detailing. The following conclusions are drawn from this study:

1. The seismic zone and soil type together determine the design spectral acceleration — site-specific ground response studies are strongly recommended for buildings in Zones IV and V.
2. Structural regularity — both in plan and elevation — is the single most important configuration decision. Irregular buildings consistently perform worse in earthquakes and require more complex analysis.
3. The SMRF with ductile detailing per IS 13920:2016 provides the minimum acceptable performance for buildings in Zone III and above. OMRF frames are only permitted in Zone II.
4. The "strong column — weak beam" hierarchy is non-negotiable. Column failure leads to pancake collapse; beam hinge mechanisms allow controlled energy dissipation with life-safety preservation.
5. Advanced technologies (base isolation, viscous dampers, BRBFs) offer significant performance advantages for critical facilities and high seismic zones, and their cost premium is justified by reduced downtime, damage, and life-cycle costs.
6. Drift control ($\leq 0.4\%$ storey height) governs the design of taller buildings more often than strength, making stiffness a primary design parameter.
7. Quality of construction and supervision is equally important as design. Even a perfectly designed structure will fail if bar bending, concrete placement, and cover are not properly executed on site.

References

IS 1893:2016 Part 1	Criteria for Earthquake Resistant Design of Structures — General Provisions and Buildings
IS 13920:2016	Ductile Design and Detailing of RC Structures Subjected to Seismic Forces
IS 456:2000	Plain and Reinforced Concrete — Code of Practice
IS 875 Part 1/2/3	Code of Practice for Design Loads — Dead, Live, and Wind Loads
FEMA 450 (NEHRP)	Recommended Provisions for Seismic Regulations for New Buildings
Chopra, A.K.	Dynamics of Structures — Theory and Applications to Earthquake Engineering, 4th ed.

Report prepared by Sunny Lakhwani — CodeAlpha Civil & Structural Engineering Internship | Task 4